

Desigualdad de Minkowski

Proposición 1 (Desigualdad de Minkowski). Sea $1 \leq p \leq \infty$ y $f, g \in \mathcal{L}^p(\mu)$

$$\implies \|f + g\|_p \leq \|f\|_p + \|g\|_p.$$

Demostración: Dividimos la demostración en tres casos excluyentes:

I. Si $p = 1$, entonces podemos aplicar directamente la desigualdad triangular de $|\cdot|$:

$$\begin{aligned} \|f + g\|_1 &= \int_{\Omega} |f(x) + g(x)| \, d\mu(x) \stackrel{\Delta}{\leq} \int_{\Omega} [|f| + |g|] \, d\mu(x) \\ &\stackrel{\text{lineal}}{=} \int_{\Omega} |f| \, d\mu(x) + \int_{\Omega} |g| \, d\mu(x) = \|f\|_1 + \|g\|_1. \end{aligned}$$

II. Si $1 < p < \infty$, entonces

$$\begin{aligned} |f(x) + g(x)|^p &= |f(x) + g(x)| |f(x) + g(x)|^{p-1} \\ &= |f(x)| |f(x) + g(x)|^{p-1} + |g(x)| |f(x) + g(x)|^{p-1}. \end{aligned} \tag{1}$$

Integrando el primer sumando del lado derecho, obtenemos

$$\begin{aligned} \int_{\Omega} |f(x)| |f(x) + g(x)|^{p-1} \, d\mu(x) &\stackrel{\text{Hölder}}{\leq} \|f\|_p \| |f + g|^{p-1} \|_q = \\ &\leq \|f\|_p \left[\int_{\Omega} |f + g|^{(p-1)q} \, dP \right]^{1/q} = \\ &\leq \|f\|_p \left[\int_{\Omega} |f + g|^p \, dP \right]^{\frac{1}{p} \cdot \frac{p}{q}} = \|f\|_p \left(\|f + g\|_p \right)^{p/q}. \end{aligned} \tag{2}$$

Entonces, como consecuencia de (1) y (2), deducimos que

$$\begin{aligned} \|f + g\|_p^p &\stackrel{(1)}{=} \int_{\Omega} |f(x)| |f(x) + g(x)|^{p-1} \, d\mu(x) + \int_{\Omega} |g(x)| |f(x) + g(x)|^{p-1} \, d\mu(x) \\ &\stackrel{(2)}{\leq} \|f\|_p \|f + g\|_p^{p/q} + \|g\|_p \|f + g\|_p^{p/q} = \|f + g\|_p^{p/q} [\|f\|_p + \|g\|_p] \cdot \\ \iff \|f + g\|_p^{p-p/q} &= \|f + g\|_p \leq \|f\|_p + \|g\|_p. \end{aligned}$$

III. Si $p = \infty$, entonces tomamos

$$A_1 \subset \Omega : \|f\|_\infty = \sup_{x \in A_1^c} |f(x)| \quad \text{y} \quad A_2 \subset \Omega : \|g\|_\infty = \sup_{x \in A_2^c} |g(x)|.$$

Definimos $A := A_1 \cup A_2$, entonces

$$\begin{aligned} \|f\|_\infty + \|g\|_\infty &= \sup_{x \in A_1^c} |f(x)| + \sup_{x \in A_2^c} |g(x)| \geq \sup_{x \in A^c} |f(x)| + \sup_{x \in A^c} |g(x)| \\ &\geq \sup_{x \in A^c} [|f(x)| + |g(x)|] \underset{\Delta}{\geq} \sup_{x \in A^c} |f(x) + g(x)| \\ &\geq \inf_{\substack{B \in \Sigma \\ \mu(B)=0}} \sup_{x \in B^c} |f(x) + g(x)| = \|f + g\|_\infty. \end{aligned}$$

Como se ha demostrado en los tres casos, se tiene que $\|f + g\|_p \leq \|f\|_p + \|g\|_p$. ■

Referenciado en

- Ley-debil-grandes-numeros
- Lem-esp-lp-vectorial
- Lem-esp-lp-normado